

Additional results

Companion volume to *preliminary results*: deeper descriptive cuts, per-firm decompositions, and market-structure context

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*Extended robustness and other findings to **preliminary results**: Spec C extensions (Cogen / Biomass / Nuclear in §??; per-firm $\times$ tech in §??), plus findings outside the headline memo's scope (price-setting, geographic CCGT dominance, post-DA intervention chain, cost cascade). All identification claims use the same methodology (Spec A BSTS, Spec B per-hour-class BSTS, Spec C per-curve DiD), same window-and-market-specific bandwidth h ($p_{90} - p_{50}$ of MCP per cell; preliminary results §1), same 2024 placebo. Seasonality handled by BSTS internally plus same-calendar matching; per-curve metrics at the wider $h = 140$ are descriptive supplements only.*

Contents

Glossary of acronyms

CCGT combined-cycle gas turbine — **DA** day-ahead market — **IDA** intraday auctions (3 European SIDC sessions post-2024-06-14) — **MCP** market clearing price — **MTU** market time unit (MTU60 hourly, MTU15 quarter-hourly) — **ISP15** Imbalance Settlement Period at 15 min (started 2024-12-01) — **REE** Red Eléctrica de España (TSO) — **OMIE** wholesale market operator — **PDBC** OMIE DA-cleared schedule — **PDBF** PDBC plus bilaterals — **PHF** post-IDA + post-RT final schedule — **TR** Tiempo Real real-time restrictions (PO 3.4–3.5, pay-as-bid) — **Fase I/II** pre-IDA / post-IDA technical restrictions (pay-as-bid) — **aFRR** / **mFRR** / **RR** balancing reserves (PICASSO / MARI / TERRE, marginal) — **BSP** Balance Service Provider — **EUPHEMIA** DA pan-European auction algorithm.

Reform-window calendar and confounders

We use the same five-regime reform-window timeline as the headline memo (*preliminary results* §1, Figure 1). Two confounders to flag for any cross-regime reading in this document.

Calendar non-overlap. The five regime windows cover different calendar months (3-sess Jun–Nov 2024; ISP15-win Dec 2024–Mar 2025; MTU15-IDA pre-blk late Mar–Apr 2025; MTU15-IDA post-blk May–Sep 2025; DA15/ID15 Oct 2025 onwards). Spanish electricity is strongly seasonal, so raw cross-regime means mix the regime effect with the seasonal mix. Three regimes (3-sess, ISP15-win, MTU15-IDA pre-blk) have a clean same-calendar pre-IDA baseline available (Jun–Nov, Dec–Mar, late-Mar–Apr in 2018–2023, none under any granularity reform); the post-blk and DA15/ID15 windows are reforzada-confounded with no clean granularity-only same-cal comparator. *Identification claims defer to the headline memo's BSTS + 2024 same-calendar placebo, where seasonality is absorbed by construction.*

Cross-border co-treatment for ID15. MTU15-DA was an EU-wide “Big Bang” on 2025-09-30: all SDAC bidding zones cleared their first 15-min day-ahead simultaneously, so cross-border DA allocation changed identically for Spain and its neighbours — a critical/flat DiD on cross-border net flow at the cutover is a clean null ($\theta = 793$ MW, $t = 0.73$). MTU15-IDA is different: it cut the bid time unit of the already-pan-European SIDC and XBID venues from 60 to 15 min, so cross-border *allocation timing* co-treats with bid-granularity — the same DiD on net flow returns $\theta = 8965$ MW ($t = 5.12$), persistent across post-window truncations. The DA15 cross-border channel is therefore empirically inert; any ID15 headline reads as bundling bid-granularity with cross-border-allocation-timing within the same intervention.¹

BSTS / CausalImpact: model, math and configurations

Every Spec A result in this document — and every BSTS effect cited from *preliminary results* §4.A — is produced by `CausalImpact::CausalImpact` (Brodersen *et al.* 2015), an R wrapper around the Bayesian structural time-series engine `bsts` of Scott & Varian (2014). The model is a linear-Gaussian state-space form fit by Markov-chain Monte Carlo on the pre-intervention window and projected as a posterior counterfactual into the post-intervention window.

Notation and structural form

Let $y_t \in \mathbb{R}$ be the outcome at time $t = 1, \dots, T$, and $x_t \in \mathbb{R}^p$ a vector of contemporaneously-observed covariates. The intervention time T_0 partitions the index into a pre-window $\mathcal{T}_0 = \{1, \dots, T_0\}$ and a post-window $\mathcal{T}_1 = \{T_0+1, \dots, T\}$. The data-generating process for y_t decomposes into a trend, K seasonal cycles, a contemporaneous regression, and Gaussian observation noise:

$$y_t = \mu_t + \sum_{k=1}^K S_t^{(k)} + \beta^\top x_t + \varepsilon_t, \quad \varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_y^2). \quad (1)$$

Components

Local level (trend). The trend is a random walk *without* drift slope:

$$\mu_{t+1} = \mu_t + \eta_t^\mu, \quad \eta_t^\mu \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\mu^2). \quad (2)$$

The local linear trend variant (slope state ν_t , $\mu_{t+1} = \mu_t + \nu_t + \eta_t^\mu$, $\nu_{t+1} = \nu_t + \eta_t^\nu$) is rejected: a slope state projects linearly into $\mathcal{T}_1$, extrapolating years of pre-window growth into the post-window. The local level lets the underlying y -level wander without imposing a direction.

Dummy seasonal of integer period S . Constrained-sum recursion:

$$S_t^{(\text{dum}, S)} = - \sum_{j=1}^{S-1} S_{t-j}^{(\text{dum}, S)} + \omega_t^{(S)}, \quad \omega_t^{(S)} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_S^2). \quad (3)$$

The state vector holds the last $S-1$ values of $S_t^{(\text{dum}, S)}$; the constraint $\sum_{j=0}^{S-1} S_{t-j}^{(\text{dum}, S)}$ is small (only ω_t -wide), so the seasonal averages to zero over each full cycle. Standard choices: weekly ($S = 7$) and diurnal ($S = 24$).

¹SDAC 15-minute MTU Information Package (<https://www.nemo-committee.eu/assets/files/sdac-15-minute-mtu-information-paper.pdf>). Cross-border flows: ESIO indicators 535 (FR import), 536 (FR export), 539 (PT import), 540 (PT export); net flow per (date, clock-hour) computed as $(535 + 539) - (536 + 540)$ summed across the 4 quarters of the hour.

Trigonometric (Harvey) seasonal of period P and F harmonics. For each harmonic $f \in \{1, \dots, F\}$ with frequency $\lambda_f = 2\pi f/P$, define a pair of states $(\gamma_{f,t}, \gamma_{f,t}^*) \in \mathbb{R}^2$ that rotates with innovation:

$$\begin{pmatrix} \gamma_{f,t+1} \\ \gamma_{f,t+1}^* \end{pmatrix} = \begin{pmatrix} \cos \lambda_f & \sin \lambda_f \\ -\sin \lambda_f & \cos \lambda_f \end{pmatrix} \begin{pmatrix} \gamma_{f,t} \\ \gamma_{f,t}^* \end{pmatrix} + \begin{pmatrix} \eta_{f,t} \\ \eta_{f,t}^* \end{pmatrix}, \quad \eta_{f,t}, \eta_{f,t}^* \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_{\text{trig}}^2). \quad (4)$$

The contribution to y_t is the sum of the “real” components: $S_t^{(\text{trig}, P, F)} = \sum_{f=1}^F \gamma_{f,t}$. The block costs $2F$ states regardless of P . For an annual cycle on daily data ($P = 365.25$) with $F = 2$, this is 4 states, against 364 for the dummy form.

Regression with spike-and-slab. The covariate vector enters linearly, with each coefficient carrying a binary inclusion indicator $\rho_j \in \{0, 1\}$ and a Gaussian slab:

$$\beta_j = \rho_j \tilde{\beta}_j, \quad \rho_j \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\pi), \quad \tilde{\beta} \mid \rho \sim \mathcal{N}_{|\rho|}(b_\rho, V_\rho), \quad (5)$$

with $|\rho| = \sum_j \rho_j$. The slab covariance V_ρ is a Zellner-style g -prior calibrated to $X^\top X$; the prior inclusion probability is $\pi = E[m]/p$ with default expected model size $E[m] = 3$.

State-space canonical form

Collecting the dynamic components into a state vector $\alpha_t \in \mathbb{R}^d$ (local level + dummy seasonals + trig pairs, with β held separately as a static parameter), the model is the linear-Gaussian state-space system

$$\alpha_{t+1} = T \alpha_t + R \eta_t, \quad \eta_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, Q), \quad (6)$$

$$y_t = Z \alpha_t + \beta^\top x_t + \varepsilon_t, \quad \varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_y^2), \quad (7)$$

where the transition matrix T is block-diagonal,

$$T = \text{blockdiag} \left(\underbrace{1}_{\text{level}}, \underbrace{T_{\text{dum}, S}}_{(S-1) \times (S-1) \text{ companion}}, \underbrace{T_{\text{trig}, F, P}}_{2F \times 2F \text{ block-rotation in (??)}} \right),$$

R is the selector matrix that picks out the components receiving innovations, Z is the time-invariant observation row that extracts μ_t , the leading dummy-seasonal slot, and $\sum_f \gamma_{f,t}$, and the innovation covariance is $Q = \text{diag}(\sigma_\mu^2, \sigma_S^2 \mathbf{1}_{S-1}, \sigma_{\text{trig}}^2 \mathbf{1}_{2F})$.

Priors

All innovation variances and the observation variance carry conjugate inverse-Gamma priors:

$$\sigma_\mu^2, \sigma_S^2, \sigma_{\text{trig}}^2, \sigma_y^2 \stackrel{\text{indep}}{\sim} \text{InvGamma}(\nu_\bullet/2, s_\bullet/2),$$

with default prior degrees of freedom $\nu_\bullet \approx 1$ and prior scales $s_\bullet$ calibrated to the empirical variance of y . The regression prior is (??). All priors are mildly informative; at the daily/hourly precision used here, the posterior is dominated by the likelihood.

Posterior inference (Gibbs sampling)

The joint posterior $p(\alpha_{1:T_0}, \theta \mid y_{1:T_0})$, with $\theta = (\sigma_\mu^2, \sigma_S^2, \sigma_{\text{trig}}^2, \sigma_y^2, \beta, \rho)$, has no closed form. We sample from it via a two-block Gibbs scheme:

- (a) *States given parameters.* For fixed θ , draw $\alpha_{1:T_0} \sim p(\cdot \mid y_{1:T_0}, \theta)$ via the Durbin–Koopman simulation smoother: forward Kalman filter, backward simulation pass.

- (b) *Parameters given states.* Conditional on $\alpha_{1:T_0}$, the innovation variances draw from inverse-Gamma conjugate posteriors; (β, ρ) draws from a stochastic-search variable-selection (SSVS) update that integrates out the slab and toggles each ρ_j by a Bernoulli step.

We run `niter` = 2000 iterations with the default 10% burn-in; convergence is monitored via the effective sample size reported by `bsts`.

Counterfactual and treatment-effect summaries

The post-window counterfactual is the posterior predictive of y_t under the trained model evaluated at the *actually-observed* post-window covariates x_t (no extrapolation):

$$\hat{y}_t^{(0)} \sim p(y_t \mid y_{1:T_0}, x_t), \quad t \in \mathcal{T}_1.$$

The pointwise treatment effect is $\delta_t = y_t - \hat{y}_t^{(0)}$. We report the average effect

$$\bar{\delta} = \frac{1}{|\mathcal{T}_1|} \sum_{t \in \mathcal{T}_1} \delta_t,$$

its 95% credible interval, and the Bayesian tail $\Pr(\bar{\delta} \leq 0 \mid y_{1:T})$ (the BSTS “p-value”, a posterior tail, not a frequentist test).

What is extrapolated, and the robustness check for what isn’t. In the post-window the local level μ_t is held flat at μ_{T_0} ; the dummy seasonal cycles continue forward at their pre-window-fitted shapes; the regression contribution $\beta^\top x_t$ is evaluated at the *actually-observed* post-window covariates. Annual seasonality therefore enters the counterfactual primarily through the covariate block: wind, solar and gas all carry strong annual cycles and their post-window realisations are observed — so cross-month level shifts (winter heating, summer cooling, solar amplitude) are carried by the regression term, not by extrapolating any seasonal state. Residual annual variation that (wind, solar, gas) do not span is the one channel exposed to bias when the pre and post windows fall on different calendar bands. The placebo BSTS at fake cutover 2024-10-01 (*preliminary results* §4.A.6) is the robustness check for this residual leakage — a non-zero placebo $\bar{\delta}$ would wound the corresponding real estimate. The multi-cycle extension above is held in reserve as the heavier remedy if the placebo were to flag something.

Cycle stack: baseline and proposed extension

Baseline (current implementation). Local level + one dummy seasonal + spike-and-slab regression on (wind, solar, gas). The seasonal block is weekly ($S = 7$, 6 states) on daily data, or diurnal ($S = 24$, 23 states) on hourly data. The state dimension is small ($d \in [7, 24]$) and the one-line `model.args` call to `CausalImpact` accommodates it. Low-frequency cross-month variation that the weekly/diurnal cycle cannot absorb is left to the regression block on (wind, solar, gas).

Proposed multi-cycle extension (not yet run). Three seasonal layers plus the local level, to disentangle short-, medium- and low-frequency seasonality explicitly:

Layer	Form	Configuration	States
Diurnal (hourly data only)	dummy	$S = 24$, one obs per slot	23
Weekly	dummy	$S = 7$, one day (or 24 hours) per slot	6
Annual	trig.	$P = 365.25$ d (or 8766 hr), $F = 2$ harmonics	4

The annual trig layer absorbs the systematic Apr–Sep vs Oct–Mar level differences in Spanish electricity (winter heating, summer cooling, hydro inflows, solar amplitude) that the baseline weekly cycle cannot reach and that the existing implementation leaves to the regression block. Adopting it requires constructing the `bsts` state specification manually — `AddLocalLevel` → `AddSeasonal` → `AddTrig` — and passing the resulting model to `CausalImpact` via the `bsts.model` interface; the existing one-line `model.args` call exposes only a single seasonal layer.

Why trigonometric, not dummy, for the annual layer. A 365-slot dummy seasonal needs three full years of pre-window before each slot has even three independent draws; identification is poor and MCMC wastes iterations on slots the data barely informs. The trigonometric form imposes smooth low-rank structure at $2F$ states regardless of period P . The harmonic count F controls bias–variance: $F = 1$ gives a single sinusoid; $F = 2$ admits a winter *and* a summer peak with an autumn dip — the natural shape for Spanish electricity, where December heating and July–August cooling are both demand peaks.

Pre/post windows

MTU15-IDA: pre 2022-01-01 to 2025-03-18, post 2025-03-19 to 2025-04-27 (40 days, pre-blackout, reforzada-absent). **MTU15-DA α :** pre 2025-04-28 to 2025-09-30 (5 months, post-reforzada-onset, pre-MTU15-DA), post 2025-10-01 to 2025-12-31 — separates granularity from reforzada. **MTU15-DA β :** pre 2022-01-01 to 2025-09-30 (long pre, joint granularity + reforzada). The spatial blackout-cutover BSTS in §?? uses pre 2024-06-14 to 2025-04-27, post 2025-04-28 to 2026-01-31.

Why BSTS over a panel DiD for level outcomes

The DA price and per-tech cleared MWh series are strongly non-stationary (Spanish renewable capacity grew $\sim 6\times$ over 2018–2025) and carry a multi-scale seasonal structure. A panel DiD on a single outcome series cannot absorb all the cycles plus the regression transparently. BSTS decomposes the variation explicitly via (§?), lets the spike-and-slab select among candidate covariates, and produces a counterfactual conditioned on the actually-realised post-period exogenous variables — not on out-of-sample extrapolation. The DiD framework is reserved here for the within-day redistribution objects (Spec C σ_p , N_{eff}), where critical and flat curves clear within the same day and the comparison absorbs all between-day variation by construction — a different identification logic where BSTS would lose precision.

Reading guide. Part A is about bidding and trading patterns. Part B is about the regulatory/post-clearing channels (reforzada, geographic dominance, CNMC sanctions) that are independent of the granularity reform but constrain how the identification can be set up.

Part I

Part A — Bidding and trading patterns

Reading guide for Part A. Three setup sections come first. §?? defines the **hour-class taxonomy** (critical / flat / midday) used to slice every table and figure that follows. §?? characterises **who price-sets** in the day-ahead market (§??) and, methodologically, in the intraday markets (§??). §?? reports the bid-shape metrics (σ_p , N_{eff} — the same per-curve functionals as the headline memo *preliminary results* §1, §4.C). Quantitative tables come in §??; the Spec C extensions at the headline-memo bandwidth follow in §?? (per-firm $\times$ tech) and §?? (tech-scope robustness).

1 Hour-class taxonomy — supplementary classifier table

The critical/flat partition is defined and motivated in *preliminary results* §2 (which uses the morning-ramp $\cup$ evening-ramp union as “critical”, overnight hours as “flat”, midday as a separate falsification class). This subsection only supplies the full per-hour classifier table that is too long for the headline memo, plus two validation checks.

Table 1: Hour-of-day classification metrics, Spain 2023–2026 (ENTSO-E A65 + A75, 15-min ISP, $n \approx 3.4$ years of daily observations per hour). The classifier is σ_{within} , the expected SD of residual demand (load – wind – solar) across the 4 quarters of each clock-hour. Critical hours: $\sigma_{\text{within}} \in [522, 1303]$. Flat hours: $\sigma_{\text{within}} \in [336, 357]$. Midday: $\sigma_{\text{within}} \in [468, 502]$ (intermediate; excluded from main specs because of the solar-saturation channel). Dropped (transition) hours sit at the regime boundaries. *Year-stability*: across 2023, 2024, 2025 the critical-set min σ_{within} exceeds the flat-set max by $1.4\times$; no within-class swaps across years.

Clock-hour	Demand (GW)	Solar (GW)	Wind (GW)	Residual demand (GW)	σ_{within} (MW)	CV (%)	Δ demand (MW)	Class
00:00–01:00	22.8	0.1	7.3	15.4	412	2.7	–834	<i>dropped</i>
01:00–02:00	22.0	0.1	7.1	14.8	347	2.4	–470	<i>flat</i>
02:00–03:00	21.6	0.1	6.9	14.6	336	2.3	–190	<i>flat</i>
03:00–04:00	21.5	0.1	6.7	14.7	357	2.4	176	<i>flat</i>
04:00–05:00	22.4	0.1	6.7	15.7	587	3.8	1114	<i>dropped</i>
05:00–06:00	24.4	0.2	6.7	17.5	721	4.1	1743	<i>critical</i>
06:00–07:00	26.5	1.5	6.6	18.4	915	5.0	1423	<i>critical</i>
07:00–08:00	27.7	4.9	6.2	16.6	1078	6.5	627	<i>critical</i>
08:00–09:00	28.3	9.1	5.8	13.4	1006	7.5	253	<i>critical</i>
09:00–10:00	28.3	12.0	5.5	10.9	753	6.9	–104	<i>dropped</i>
10:00–11:00	28.2	13.3	5.4	9.5	499	5.3	–66	<i>dropped</i>
11:00–12:00	28.2	13.8	5.5	8.9	468	5.3	94	<i>dropped</i>
12:00–13:00	28.3	13.8	5.7	8.8	502	5.7	–104	<i>dropped</i>
13:00–14:00	27.9	13.5	5.8	8.5	485	5.7	–331	<i>dropped</i>
14:00–15:00	27.5	12.9	6.1	8.5	469	5.5	–122	<i>dropped</i>
15:00–16:00	27.6	11.7	6.4	9.6	677	7.1	291	<i>dropped</i>
16:00–17:00	28.2	9.0	6.8	12.3	1109	9.0	553	<i>critical</i>
17:00–18:00	29.2	5.4	7.4	16.4	1303	7.9	891	<i>critical</i>
18:00–19:00	30.5	2.2	7.7	20.5	1065	5.2	884	<i>critical</i>
19:00–20:00	31.3	0.6	7.9	22.8	522	2.3	176	<i>critical</i>
20:00–21:00	30.4	0.3	7.9	22.2	622	2.8	–1444	<i>critical</i>
21:00–22:00	28.0	0.2	7.9	20.0	781	3.9	–1851	<i>critical</i>
22:00–23:00	25.8	0.2	7.7	18.0	664	3.7	–1407	<i>critical</i>
23:00–00:00	24.2	0.1	7.5	16.5	510	3.1	–1062	<i>dropped</i>

Two validation checks. (i) *No cost-shock confound*: CCGT price-setting share is 9.1% critical vs 9.0% flat — gas-cost shocks cannot propagate asymmetrically across the two classes, so within-day differencing identifies behaviour, not cost. (ii) *Outcome-side*: post-MTU15-DA the within-hour DA quarter clearing-price SD averages 6.58 EUR/MWh in critical hours vs 2.13 in flat ($3.1\times$), so the classifier correctly predicts where post-reform quarter prices actually diverge.

2 Who clears in which hours: the tech-mix by hour-class

Before turning to who price-sets, it is worth fixing the basic descriptive picture: in each hour-class, what share of the final cleared MWh actually comes from each technology? The cross-tab in Table ?? is computed on the *final program* (PHF, post-IDA) for the post-DA15 window (2025-10-01 to 2025-12-31), so it answers the natural question “after every market has cleared, who is actually delivering the energy in each hour-class?”

Tech	Morning ramp (5–8)	Evening ramp (16–22)	Midday (11–14)	Flat (1–3)
Nuclear	34.3	26.9	25.8	34.0
Wind	20.9	16.3	13.7	21.6
CCGT	14.0	13.2	9.2	13.6
<i>Unknown</i> [†]	12.1	13.3	14.7	12.8
Hydro	10.0	13.4	4.6	9.7
Cogen	5.1	4.0	3.5	5.1
Hydro_RES	1.4	1.2	1.0	1.4
Biomass	1.2	0.9	0.9	1.2
Hydro_pump	0.9	1.9	0.1	0.5
Solar PV	0.0	8.7	26.4	0.0
Coal	0.1	0.1	0.1	0.1
Hour-class total (TWh)	13.9	31.1	18.5	10.5

Table 2: Final-program (PHF) share of cleared MWh by tech and hour-class, post-DA15 window. Window 2025-10-01 to 2025-12-31 (92 days). Numbers are % of total cleared MWh in each hour-class (column). PHF is the post-IDA program; we take the last-session assigned MW per (date, period, unit). The hour-class totals (TWh) reflect different hour counts per class per day (4 morning-ramp, 7 evening-ramp, 4 midday, 3 flat). [†]“Unknown” is the ~13% of cleared MWh from unit codes not in the unit-map (mostly HISVD*/GALVD*/GNVD*/GSVD* virtual-delivery families — portfolio/aggregator units that the unit-map drops); not a tech category. Reproduction: `scripts/analysis/firm/tech_mix_by_hour_class.py`.

Reading the cross-tab.

- **Nuclear is the largest single bucket in every non-midday hour-class** (~34% morning ramp and flat, ~27% evening). Inframarginal baseload with no within-hour variation — it sets the floor of the merit order in every hour.
- **Solar PV** is the headline midday tech (26.4% midday, ~0% overnight, 8.7% in the evening ramp where the ramp extends past sunset). The diurnal cycle is fully visible.
- **Wind** is the second-largest in every hour-class except midday (~20% morning and flat), dropping to ~14% midday when solar takes share.
- **CCGT** (the strategic-bidder pool of *preliminary results* §4.C) clears ~13–14% of total in every non-midday hour-class — sizeable but not dominant. Drops to ~9% in midday when solar pushes it off the merit order. The strategic-engagement story of *preliminary results* §4.C is therefore about restructuring the bid-shape of a ~13%-of-cleared-MWh fleet, not about reshuffling the merit order at any margin.
- **Cogen** (separate from CCGT, see below) clears ~3–5% in every hour-class. *Its share is roughly flat across hour-classes*, consistent with the heat-driven (non- strategic) dispatch story — Cogen doesn’t shift to follow the power-side demand profile because its operating decision is driven by industrial heat demand and a regulated remuneration regime, not by the wholesale price.
- **Hydro** is meaningfully higher in critical hours (evening 13.4%, morning 10%) than in midday (4.6%), consistent with the reservoir-arbitrage story (dispatch into peaks).
- **Hydro_pump** is small everywhere but skews toward critical evening (1.9% vs 0.1% mid-day) — discharges into the evening peak, near-zero in midday when it would be pumping.

Why CCGT and Cogen are kept separate. The unit-map distinguishes CCGT (58 large units, Big-4 dominated, two-tier strategic bid curve with Fase I-recall economics) from Cogen (434 small units, regulated heat-driven dispatch under Real Decreto 413/2014 — the “~5% dispatchable tail” of *preliminary results* App. C.1 footnote). Pooling would dilute the strategic-engagement story; the analysis throughout restricts to CCGT.

Cogen as a Spec C null: a tech-scope robustness check

Spec C in *preliminary results* §4.C runs on CCGT, hydro, pump-storage and wind. Cogen is excluded *ex ante* on a regulated-remuneration prior: heat-driven dispatch should remove the within-hour optimisation problem that the granularity reform expands. We verify the prior by running the same Spec C on Cogen at the same bandwidth ($h = 50$ EUR/MWh) and windows (pre 2025-04-28 to 2025-09-30, post 2025-10-01 to 2025-12-31); Biomass and Nuclear reported alongside.

Table 3: Spec C robustness on technologies excluded from *preliminary results* §4.C. Per-curve DiD of σ_p and N_{eff} at the same DA15 DA bandwidth ($h = 50$ EUR/MWh) and windows (pre 2025-04-28 to 2025-09-30; post 2025-10-01 to 2025-12-31) used in the headline memo. DiD = (post crit – post flat) – (pre crit – pre flat). CCGT row added for orientation to the headline memo’s reported value (*preliminary results* §4.C Table: CCGT DA σ_p DiD 0.68, N_{eff} DiD 1.89). Per-cell sample sizes in the source CSV. Reproduction: `scripts/analysis/firm/cogen_stability_check.py`.

Tech	σ_p (EUR/MWh)		DiD σ_p	N_{eff}		DiD N_{eff}
	post (crit, flat)	pre (crit, flat)		post (crit, flat)	pre (crit, flat)	
CCGT (in scope)	3.14, 2.18	2.55, 2.39	0.79	3.85, 2.40	2.85, 3.39	1.99
Cogen	0.71, 1.03	0.49, 0.74	−0.08	1.04, 1.06	1.04, 1.05	−0.00
Biomass	1.74, 1.95	0.82, 0.69	−0.34	1.57, 1.69	1.32, 1.35	−0.09
Nuclear	0.59, 0.55	1.83, 3.30	1.51	1.65, 1.57	1.80, 1.74	0.03

Reading. Cogen σ_p DiD −0.08, N_{eff} DiD ≈ 0 — the prior holds. Biomass behaves like Cogen (second regulated-remuneration tech, no granularity response). Nuclear’s 1.51 comes from a thin pre-flat cell ($n = 233$, 4 units) and is not interpreted. CCGT reproduces *preliminary results* §4.C within rounding (0.79 vs published 0.68***; 1.99 vs 1.89***; unweighted-mean vs unit-FE coefficient).

Quantity-side mirror. PHF cleared MWh/day stays flat for Cogen across all six regimes (Cogen 35.1 / 42.6 / 44.1 / 34.0 / 41.4 / 44.3 GWh/day for pre-IDA / 3-sess / ISP15-win / MTU15-IDA pre-blk / MTU15-IDA post-blk / DA15/ID15), against CCGT tripling (39 → 129). Biomass similarly flat (~ 10 GWh/day). Cogen operates under the regulated renewables-and-cogen remuneration framework (Real Decreto 413/2014): heat-driven dispatch, single-block must-clear bid at ~ 51 /MWh, no within-hour optimisation problem — so granularity adds no instrument and produces no DiD signal.

3 Who price-sets

3.1 For the day-ahead market

Method. The price-setter is the unit whose at-MCP step is *partially* accepted: $q_{\text{below}} < q_{\text{assigned}} < q_{\text{below}} + q_{\text{at}}$ (with q_{at} the MW bid at MCP ± 0.01 EUR/MWh). Full acceptance (inframarginal) and curtailed-to-zero (minimum-income deactivation) excluded. Aggregations MW-weighted by q_{at} .

Who is marginal. Wind is the largest sell-side price-setter in almost every cell (18–71% of at-MCP MW), reflecting renewable aggregators landing a single block at-the-money. Hydro and pump-storage next; in DA15/ID15, Iberdrola owns $\approx 65\%$ critical and 80% flat through pump, hydro and wind. **CCGT’s share is small and volatile** (1–17% critical). Cross-check: CCGT critical/flat is near-symmetric (3-sess 8.0/8.9, DA15/ID15 2.7/2.0), so a gas-cost shock

— which hits both classes alike — is not what within-day differencing detects. Per-firm split within the CCGT pool:

Regime	Hour-class	IB	GE	GN	HC	OTHER
3-sess	Critical	60	31	0	0	8
3-sess	Flat	16	50	0	0	25
ISP15-win	Critical	35	57	0	0	5
ISP15-win	Flat	2	74	0	3	20
DA60/ID15 post-blk	Critical	31	69	0	0	0
DA60/ID15 post-blk	Flat	19	80	0	0	0
DA15/ID15	Critical	65	29	0	6	0
DA15/ID15	Flat	80	0	6	14	0

Table 4: Per-firm shares within the CCGT price-setter pool (% of CCGT q_{at}). GN’s CCGT bid stack is so heavily concentrated far above the clearing-price ceiling (a flat shelf at $\sim 1,000$ EUR/MWh) that GN essentially never lands at MCP. IB and GE alternate as the dominant CCGT price-setter. The 33.1% CCGT flat-hour figure in DA60/ID15 post-blackout is 80% GE.

Price-setting $\neq$ market power for CCGT. A unit is flagged “marginal” merely for having offered MW at the price that happened to clear. The ranking actually *inverts* the market-power ordering: GN withholds the most (75% parked, §??) but scores 0% in the price-setter pool precisely *because* it bids too high ever to be marginal; IB tops the pool only by keeping a thin competing tier near MCP. CCGT market power is exercised by withholding *from* the day-ahead and captured downstream in pre-IDA Fase I redispatch (§??); we read CCGT conduct from bid-vs-marginal-cost (§??) and the Fase I channel, not from the price-setter share.

3.2 For the intraday markets

The same partial-acceptance rule, applied to the intraday-auction clearing prices, bid stack and programs (sell side, pooled across the 3 IDA sessions per day), shows **CCGT far more active as an IDA price-setter than as a DA one**: 24.2% of critical-hour IDA mass under DA15/ID15 against 2.7% in DA — a factor of ~ 9 . This is the price-setter-side trace of the day-ahead-to-intraday substitution documented in §??: CCGTs bid above the day-ahead clearing price but compete in the intraday auction, where they clear at the marginal step. Wind still dominates ($\sim 32\%$ critical) but by a smaller margin than in DA.

4 Bid-curve shape: σ_p and N_{eff}

We use the same two per-curve bid-shape scalars as the headline memo: σ_p , the MW-weighted SD of in-band tranche prices, and N_{eff} , the inverse Herfindahl of in-band tranche MW shares. Both are defined per single bid curve — one value per (unit, date, period) sell curve — with no comparison between curves. Cross-regime, cross-firm and cross-hour reading is then a comparison of the *values* σ_p and N_{eff} across curves.

4.1 Definitions and bandwidth

Bandwidth h : same as the headline memo. Both metrics restrict to a price band $|p - \text{MCP}| \leq h$ — the region where a bid is in contention. **The bandwidth is the headline memo’s** (*preliminary results* §1, Table of bandwidths): the window-and-market-specific $p_{90} - p_{50}$ of the MCP distribution in each (reform, market) cell, giving $h \in \{45, 46, 49, 50, 58, 62\}$ EUR/MWh across the eight (reform, real-or-placebo, DA-or-IDA) cells of the headline memo. This is the *strategically active competing core*: the price region in which the bid actually competes for clearance.

The MW-weighted density of $|p_{\text{bid}} - \text{MCP}|$ over all day-ahead sell tranches of the price-setting techs is **bimodal**: a dense near-MCP competing cluster (where the window-specific p_{90} bands sit) and a sparse far scarcity-cap cluster (the parked withholding tier). The two clusters meet at a sharp cliff around ~ 140 EUR/MWh, which coincides with the realised clearing-price ceiling (median MCP + 140 $\approx p_{99.9}$ of MCP). The per-curve metrics table in §?? at a single pooled $h = 140$ is reported as a *wider-band descriptive supplement*: it describes the full competitive distribution out to the cliff, useful for tech-comparison context but not for the Spec C identification. The identification reads in this document (§??, §??) all use the headline memo’s window-and-market-specific h .

The two-tier bid curve, read against marginal cost. The bimodality of Figure ?? is, unit by unit, a **two-tier bid curve**: a *competing tier* around marginal cost (clears the auction) plus a *scarcity tier* parked far above any plausible clearing price (does not clear — deliberate withholding). The parked capacity is recalled in the pre-IDA Fase I redispatch through a *separate* restriction offer (pay-as-bid), so the day-ahead scarcity-tier price is a non-payment focal choice.² The withholding boundary is the ~ 200 EUR/MWh clearing-price ceiling. Aggregated over DA15/ID15, the withheld share above the ceiling is firm-uniform in spirit but very different in magnitude: Naturgy 75% (flat shelf at $\sim 1,000$ EUR/MWh), Endesa 22% (much at the 4,000 cap), Iberdrola 22% (a 200–400 ramp), EDP-HC 19%. Iberdrola looks competitive against a 500 EUR/MWh cutoff but withholds as much as Endesa against the true ~ 200 ceiling, and clears only $\sim 16\%$ of its final programme in the day-ahead (barely above Naturgy’s $\sim 14\%$). The firms differ in *how far* they bid above the ceiling, not in *whether* they withhold. The intraday auction confirms it is strategy, not cost: on the same units and days, all four fleets bid a single competitive tier at MW-weighted 82–96 EUR/MWh with $\sim 0\%$ withheld — Naturgy at 90 EUR/MWh against its 767 day-ahead. The per-curve functionals below restrict to the competing tier, where price-graded bidding is the choice variable.

Per-curve functionals and offer-type caveat. σ_p (MW-weighted SD of in-band tranche prices) and N_{eff} (inverse Herfindahl of in-band tranche MW shares) are defined in *preliminary results* §1; we use the same definitions here. Block orders mechanically force $\sigma_p = 0$ and $N_{\text{eff}} = 1$ and MIC orders displace shaping onto the income floor, so the offer-type mix matters: wind, solar, hydro and pump-storage are $\sim 100\%$ simple; nuclear 64% simple; **CCGT is only 18% simple (50% MIC, 32% block)**. For CCGT, any geometry-to-conduct step conditions on offer type (*preliminary results* Appendix A.7 reports the type-split DiD).

4.2 Results

The results below report σ_p and N_{eff} at a wider supplementary bandwidth ($h = 140$) as cross-regime descriptive context. The Spec C identification at the headline memo’s window-and-market-specific bandwidth lives in *preliminary results* §4.C (per-tech pooled DiD), §?? (per-firm $\times$ tech), and §?? (Cogen / Biomass / Nuclear robustness).

Per-curve functionals at a wider supplementary bandwidth ($h = 140$). Table ?? and Figure ?? report σ_p and N_{eff} averaged over (unit, date, period) day-ahead sell-bid curves, by tech, hour-class and regime, at a single pooled $h = 140$ EUR/MWh — the empirical cliff between the competing cluster and the parked tier (Figure ??). *This is a wider-band supplementary view, not the Spec C identification.* The Spec C identification uses the headline memo’s window-and-market-specific bandwidth (§??) and is read in §?? (Cogen Spec C null) and §?? (per-(firm, tech)

²P.O. 3.2 process; settlement under P.O. 14.4 apartado 20 (BOE-A-2025-5342, BOE-A-2025-13076). Bidding high in DA to be excluded and then dispatched in restrictions is the conduct CNMC sanctioned in 2019 (SNC/DE/174/17, SNC/DE/175/17).

Spec C DiD); the headline DiD on the dispatchable techs lives in *preliminary results* §4.C. The wider band here adds tech-comparison context (CCGT’s competing-cluster tail extends further from MCP than hydro’s, for instance) at the cost of mixing tranches that would not be picked up by the tighter p_{90} band.

Table 5: Per-curve bid-shape metrics by tech, hour-class and regime (wider-band supplementary, $h = 140$). DA sell bids, in-band $|p - \text{MCP}| \leq 140$ EUR/MWh (the descriptive cliff bandwidth, not the headline memo’s $p_{90} - p_{50}$ bandwidth used for Spec C identification). Mean over (unit, date, period) curves. σ_p = MW-weighted price spread (EUR/MWh); N_{eff} = effective in-band tranche count. 953,384 in-band curves. Cross-regime reading: tech ranking holds, but the absolute magnitudes are inflated relative to the Spec C bandwidth by the inclusion of the outer tail of the competing cluster.

Tech	Hour	3-sess		ISP15-win		DA60/ID15 pre		DA60/ID15 post		DA15/ID15	
		σ_p	N_{eff}	σ_p	N_{eff}	σ_p	N_{eff}	σ_p	N_{eff}	σ_p	N_{eff}
CCGT	Critical	36.1	3.3	32.0	4.2	29.2	3.1	24.5	3.1	24.8	3.3
CCGT	Flat	27.5	2.5	45.2	2.7	17.1	2.3	20.2	2.6	18.2	2.3
Hydro	Critical	17.5	4.0	16.3	4.0	13.1	3.5	15.5	3.5	16.6	3.7
Hydro	Flat	17.6	4.0	18.5	3.8	12.6	3.5	16.0	3.5	15.1	3.8
Hydro pump	Critical	3.6	2.5	3.8	2.5	2.6	2.3	3.9	2.8	3.8	3.4
Hydro pump	Flat	2.1	2.6	3.9	2.9	2.4	2.3	2.6	2.4	2.8	4.1

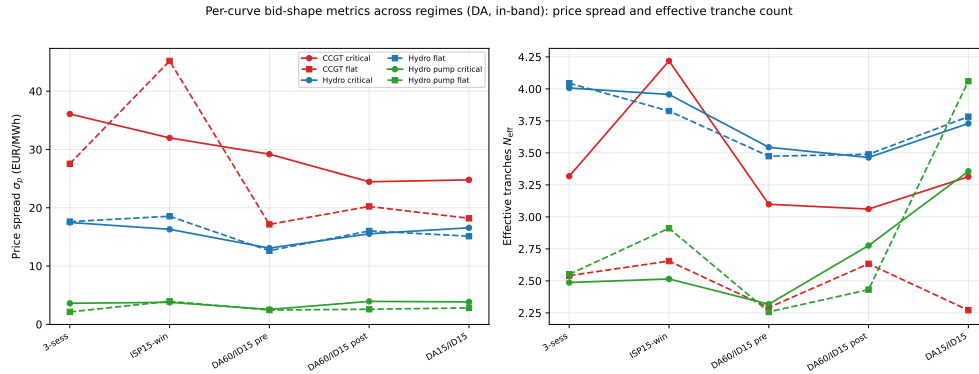


Figure 1: Per-curve bid-shape metrics across regimes (wider-band supplementary, $h = 140$). Same caveat as Table ??: wider-band descriptive view, not the Spec C identification (Spec C uses the window-specific $p_{90} - p_{50}$ bandwidth of the headline memo). Left: price spread σ_p ; right: effective tranche count N_{eff} . Each line is a (tech, hour-class) cell; solid = critical hours, dashed = flat. CCGT carries the widest price spread ($\sigma_p \approx 25\text{--}40$ EUR/MWh at this wider band) and it is larger in critical than flat hours in four of the five regimes — the qualitative pattern that Spec C identifies; Hydro is intermediate ($\sim 13\text{--}18$) and Hydro pump near-flat ($\sim 3\text{--}4$).

Wind within-hour variation is forecast, not strategy. Regressing each wind curve’s σ_p on a within-hour scarcity proxy (residual demand net of wind, deviated from clock-hour mean) with unit $\times$ date $\times$ hour FE (DA15/ID15, 2.3M in-band wind curves, date-clustered SEs): 84% of wind in-band curves are a single block and 89% of clock-hours show no across-quarter σ_p variation; where σ_p does move, the scarcity response is 0.01 EUR/MWh per GW (within-hour partial correlation 0.01) — economically negligible. The largest aggregator response, Gesternova’s $\beta = 0.14$, is still tiny against $\beta \approx 0$ for AXPO, NEXUS, IGNIS. Within-hour wind movement is the sub-hourly forecast, not strategic scarcity-chasing; wind reads as a price-taking baseline.

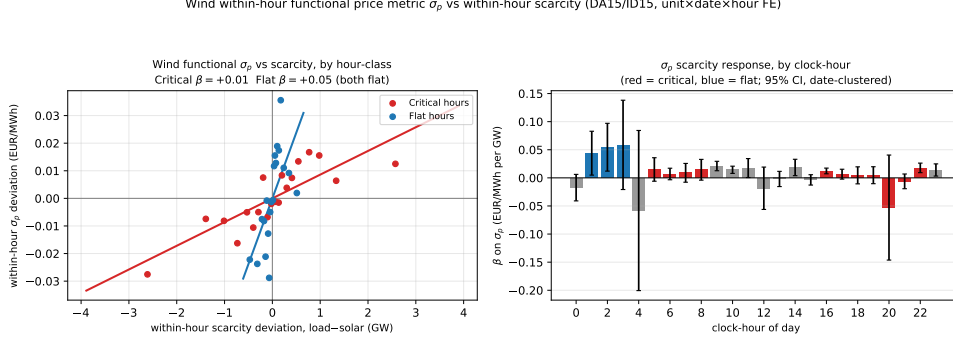


Figure 2: The wind functional price metric σ_p does not respond to within-hour scarcity (DA15/ID15, unit $\times$ date $\times$ hour fixed effects; band centred on the clock-hour-mean clearing price). Left: binned within-hour-demeaned σ_p against the within-hour-demeaned scarcity proxy (load minus solar forecast, GW), for critical and flat hours — both slopes essentially flat. Right: the slope for each clock-hour of the day (red = critical, blue = flat hours; date-clustered 95% intervals) — no hour reaches an economically meaningful response.

Pump-storage within-day arbitrage under MTU15-DA: a descriptive look (no causal claim). The bid-shape evidence above is one side of the granular-pricing reform; one would naturally look at pump-storage for the quantity-side mirror. Pre/post means on the two sides of the pump-storage book, taken at face value, show a striking within-day reallocation:

Hour-class	Generation (MW per period)		Pumping (MW per period)	
	Pre (Jul-Sep)	Post (Oct-Dec)	Pre	Post
Critical	851	1 018 (20%)	664	317 (−52%)
Flat	724	258 (−64%)	52	566 (10 $\times$)

Generation appears to reallocate from flat hours (−64%) to critical hours (20%); pumping mirrors it. Total daily generation is essentially unchanged (14 874 $\rightarrow$ 14 250 MWh, −4%). The reading would be textbook within-day arbitrage: pre-MTU15-DA, hourly bids could not price-discriminate across quarters; post-reform, the unit class with the largest within-hour arbitrage value exploits within-hour price spreads cleanly.

The headline-methodology verdict is more cautious. The BSTS counterfactual on per-tech cleared MWh (Spec A in *preliminary results* §4.A) does not retain the pump-storage MTU15-DA effect under the extrapolation-drift diagnostic (see *preliminary results* §4.A.6, “BSTS extrapolation drift cases”). The pre/post raw means above remain descriptively informative — the within-day reallocation pattern is real in the data — but we do not interpret them as a causal MTU15-DA effect for the reasons set out in the preliminary memo (cross-season pre/post, drift between the 14-day and 40-day post-windows, sensitivity to the choice of pre-window).

4.3 XBID extension of Spec C

Third-venue test of preliminary’s Spec C: σ_p and N_{eff} on Big-4 CCGT XBID sell orders (`exec_condition` $\in \{ \text{ ” }, \text{ ICE} \}$), in-band ($h = 62$ EUR/MWh) around the same-period IDA-auction clearing as anchor, around the ID15 cutover. Pre 2024-12-19 to 2025-03-18 (~ 90 days), post 2025-03-19 to 2025-04-27 (40 days, pre-blackout).

Window	Hour-class	n_{curves}	$\overline{n_{\text{tr}}}$	σ_p	N_{eff}
pre ID15	Critical	3,710	16.2	3.45	15.10
pre ID15	Flat	1,366	12.9	3.55	12.27
post ID15	Critical	2,874	13.1	2.81	12.23
post ID15	Flat	683	13.3	2.11	13.14
DiD σ_p:					0.81
DiD N_{eff}:					−3.73

Reading. Critical-vs-flat CCGT XBID σ_p widens by 0.81 EUR/MWh post-ID15 (same sign as the auction-side hydro/pump DiDs in *preliminary results* §4.C). N_{eff} tightens by −3.73 effective tranches: with the IDA auction itself moving to 15-min, CCGT XBID orders become less granular in critical hours — the continuous-market shaping load shifts to the now-finer IDA auction. The strategic-latitude story extends to a third venue, redistributively. Reproduction: `scripts/analysis/bid/xbid_specc_did.py`.

4.4 Per-firm $\times$ technology Spec C decomposition

The pooled per-(tech, market) DiD coefficients in *preliminary results* §4.C summarise the bid-shape response to MTU15-DA at the technology level. The per-firm CCGT split appears once, in *preliminary results* §A.4 (“primarily Naturgy, Iberdrola to a lesser extent, Endesa and EDP-HC do not contribute”). This subsection extends the per-firm split to all dispatchable techs in the Spec C scope (CCGT, hydro, pump-storage, wind) at the *same* per-curve outcomes (σ_p , N_{eff}), the *same* window-and-market-specific bandwidth (DA15 DA $h = 50$ EUR/MWh, DA15 IDA $h = 58$ EUR/MWh), and the *same* pre/post windows used in the headline memo. This is per-firm Spec C, not a new spec.

Table 6: Per-(firm, tech) DA15 DA Spec C DiD on σ_p and N_{eff} , $h = 50$ EUR/MWh. DiD = (σ_p post crit − post flat) − (pre crit − pre flat), computed cell-by-cell on the per-curve panel of *preliminary results* §4.C (pre 2025-07-01 to 2025-09-30; post 2025-10-01 to panel end 2025-11-09). Sign and magnitude reproduce *preliminary results* §A.4 attribution for CCGT (GN primary, IB lesser, GE near zero, HC small). Per-tech leader differs: GN on CCGT, IB and HC on hydro, GE on pump. Empty cells: firm has no units of that tech in the Spec C panel. Reproduction: `scripts/analysis/firm/per_firm_tech_bid_shape.py`.

Firm	DiD σ_p (EUR/MWh)				DiD N_{eff}			
	CCGT	Hydro	Pump	Wind	CCGT	Hydro	Pump	Wind
IB	0.96	2.44	0.24	−0.06	0.21	0.51	−0.12	−0.03
GN	3.72	1.39	0.17	0.01	7.91	0.06	0.09	−0.08
GE	−0.10	0.81	0.85	—	0.01	−0.01	0.01	—
HC	0.83	2.18	—	−0.27	0.34	0.74	—	−0.06
Headline (<i>prelim.</i> §4.C)	0.68***	1.80***	0.92***	−0.02	1.89***	0.51***	−0.06	−0.01

Reading. (i) *The CCGT per-firm pattern matches preliminary results §A.4.* GN’s σ_p DiD is 3.72 (the largest), IB’s is 0.96, HC’s 0.83, GE’s near zero. The N_{eff} DiD is even more concentrated on GN (7.91 effective tranches, vs 0.21 for IB) — GN’s competing tier restructures into much finer per-quarter ladders, GN does most of the work in the pooled CCGT 1.89 N_{eff} headline. (ii) *The per-tech leader differs by tech.* GN leads on CCGT, IB and HC on hydro, GE on pump-storage. A firm that engages with the granularity instrument on one technology does not automatically engage on every technology in its portfolio — GN’s hydro and pump σ_p DiDs are smaller than IB’s and HC’s, and GN’s N_{eff} on hydro and pump is essentially zero. (iii) *Wind is uniformly null* at the per-firm level too, consistent with the operational-uniform-across-hour-classes reading of *preliminary results* §4.C.2 (the engagement is on the volume side, not the bid-shape side). (iv) *The pooled headline rows reproduce preliminary results §4.C* within

rounding (unweighted-mean vs unit-FE regression coefficient explains the $\sim 0.1\text{--}0.2$ gap on each pooled coefficient).

Implication. Each pooled per-tech DiD of *preliminary results* §4.C is a portfolio-weighted sum of a small number of (firm, tech) cells: the CCGT σ_p widening is structurally a GN-CCGT event with small IB support; the hydro widening is IB-and-HC; the pump widening leans on GE. Each dispatchable firm responds on the technology where their portfolio gives them within-hour optimisation room.

5 Continuous-intraday repositioning (short)

XBID volume tracks the granular-auction story: gross XBID activity is $\sim 32 \rightarrow 12 \rightarrow 23$ GWh/day across the five regimes (-29% baseline to DA15/ID15, trough -62% in the 40-day MTU15-IDA-only window). **CCGT is a systematic net seller** in every regime (0.8 to 2.5 GWh/day) — the continuous-market face of “stay out of the day-ahead”. Within-hour rebalancing intensity (cross-quarter SD of XBID sell volume) is $2\text{--}5\times$ larger in critical than flat hours, ranked $\text{IB} > \text{GN} > \text{HC} > \text{GE}$, and stable across post-MTU15-IDA regimes: CI *volume* fell with the granular auction; per-firm rebalancing *intensity* did not.

Part II

Part B — Regulatory and confounder context

REE program-and-intervention chain (brief primer). Each delivery day’s unit program is built up as PDBC (DA-cleared) $\rightarrow$ PDBF (+ bilaterals) $\rightarrow$ PDVP (+ pre-IDA *Fase I/II* RT, pay-as-bid) $\rightarrow$ PHF (+ IDA + post-IDA RT2) $\rightarrow$ PHFC (+ continuous) $\rightarrow$ P48 (+ real-time TR). *Reforzada* = REE’s post-blackout (2025-04-28 onward) stance of larger pre- and post-IDA RT, mostly Fase I-up. Roadmap: q_1 drop (§??), absorbing mechanism (§??), zonal CCGT concentration (§??), system-cost view (§??), CNMC enforcement (§??).

6 The q_1 drop: CCGT auction-cleared collapse

q_1 = sum of PDBC (OMIE DA auction-cleared) MWh, restricted to Big-4 (IB, GE, GN, HC), per (technology, calendar month); same-calendar-month comparison absorbs seasonality by construction. $q_{\text{final}} = \text{PDBC} + \text{Bilaterals} + (\text{IDA} + \text{REE-RT})$. The Big-4 aggregate by tech appears in *preliminary results* Appendix C.4 and is not reproduced; the figure below adds the per-CCGT-firm split, showing the DA $\rightarrow$ RT shift is firm-wide.

Three-component decomposition (WEEKLY) per CCGT firm: DA + Bilaterals + IDA+REE-RT = q_{final} .

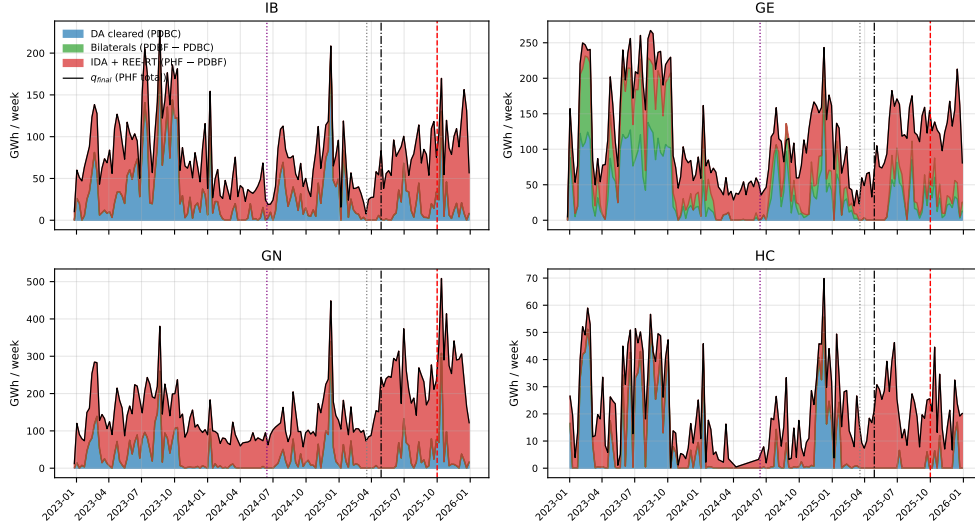


Figure 3: Weekly three-component decomposition per CCGT firm. PDBC (blue, DA-cleared) + bilateral executions (green) + IDA + REE real-time top-up (red) = final post-IDA dispatch (black), Big-4 CCGT firms split. The DA → RT shift is firm-wide and roughly tracks zonal CCGT dominance. Vertical guides: MTU15-IDA (2025-03-19), Blackout (2025-04-28), MTU15-DA (2025-10-01).

Table 7: Same-calendar-month auction-cleared DA volume by technology, GWh (PDBC, pivotal-firm aggregate). Bilaterals excluded. Each column pair compares the same calendar month one year apart, so seasonality cancels. Only CCGT collapses; every other tech grows or stays flat.

Tech	Dec 2024 → Dec 2025			Nov 2024 → Nov 2025			Aug 2024 → Aug 2025		
	Pre	Post	%	Pre	Post	%	Pre	Post	%
CCGT	1444	244	-83%	637	130	-80%	532	360	-32%
Nuclear	1253	2587	+106%	933	1909	+105%	1997	2243	+12%
Hydro	1016	640	-37%	509	611	+20%	303	424	+40%
Hydro_pump	195	247	+27%	182	341	+87%	239	406	+70%
Wind	732	696	-5%	666	1081	+62%	379	449	+18%
Solar PV	57	34	-40%	37	100	+170%	37	136	+268%

Reading. CCGT Big-4: Dec 2024 = 1,444 GWh → Dec 2025 = 244 GWh (−83%); Nov −80%. Every other tech grows or stays flat. The drop is the extensive margin (number of unit-hours cleared), not the intensive margin (per-cell mean MWh, stable at ≈ 300). Firm-wide critical-hour drops Dec24→25: IB −11%, GE −6%, GN −10%, HC −18% (flat-hour drops larger: GN −30%, HC −50%). Only CCGT can stay out of DA and rejoin via Fase I and the intraday auctions — renewables are not called up in PO 3.2 (energy-constrained), nuclear is must-clear. CCGT firms with zonal post-clearing backstop (§??) bid above DA, are rescued by pay-as-bid Fase I, and post a near-constant cap-padded DA ladder — the practice CNMC sanctioned in 2019, now at fleet-wide scale.

Reconciliation with the headline memo. Preliminary results §4.A.2 reports a CCGT DA-cleared *scale-up* of 40 GWh/day under MTU15-DA. The BSTS scale-up is the DA15 increment *conditional on a reforzada-era pre-window* (train 2025-04-28 to 2025-09-30, both pre and post post-reforzada); the −83% here is the *joint* reform-and-reforzada effect against pre-reforzada Dec 2024. Together: the reforzada onset slashed CCGT DA dramatically; within the reforzada era, DA15 partially restored it in critical hours.

6.1 Robustness: bilateral channel and outages

Bilateral (PDBF – PDBC). Could CCGT firms have relabelled output as bilateral? Table ??: for CCGT the channel is small (~ 90 – 128 GWh/month, no upward trend), so the auction-cleared collapse is real.

Table 8: Bilateral channel by technology and regime, GWh / month (PDBF – PDBC, pivotal-firm aggregate). CCGT bilaterals are small and stable; Hydro / Nuclear / Wind bilaterals are an order of magnitude larger and structurally part of their dispatch.

	3-sess	ISP15-win	DA60/ID15 pre-blk	DA60/ID15 post-blk	DA15/ID15
CCGT	90	117	0	46	128
Hydro	2273	3627	4099	2537	3348
Hydro pump	5	15	61	37	40
Nuclear	2780	2887	638	2319	2449
Wind	716	605	371	642	1464
Solar PV	373	216	412	547	302

Outages. Aggregating ENTSO-E planned + forced outage windows (PSR B04) per regime, summing ($\text{nominal}_{\text{MW}} - \text{min_avail}_{\text{MW}}$) per outage, against the ~ 23.6 GW CCGT fleet (50 of 58 units covered):

Regime	unavail MW (forced, planned)	avail. MW
pre-IDA	4 741 (815, 3 925)	18 869
3-sess	3 962 (762, 3 200)	19 648
ISP15-win	5 538 (683, 4 855)	18 072
MTU15-IDA pre-blk	5 692 (683, 5 009)	17 918
MTU15-IDA post-blk	4 895 (726, 4 170)	18 715
DA15/ID15	4 885 (2 389 , 2 495)	18 725

CCGT available capacity is essentially flat across regimes (~ 18.7 – 19.6 GW). The -83% Dec24 $\rightarrow$ Dec25 q_1 drop *cannot* be explained by reduced fleet availability. Forced outages tripled in DA15/ID15 ($726 \rightarrow 2,389$ MW) but planned outages dropped by an equal amount. ENTSO-E A73 metered output per CCGT unit (53 of 58, $\sim 91\%$ coverage) gives ~ 49 GWh/day physical generation post-DA15 against PHF-scheduled ~ 78 GWh/day — CCGT under-delivers the post-IDA program by $\sim 37\%$ via real-time TR (downward). Reproduction: `scripts/analysis/regulatory/ccgt-ou`

7 REE post-clearing inclusion as the absorbing mechanism

7.1 Full post-clearing intervention by *tipo redespacho* code

Full REE post-clearing intervention by channel (ESIOS archive 28). Labels follow project-canonical reference (parser docstring + SPANISH_MARKET_STRUCTURE.md §13). Code 23 (capacity reservation) excluded.

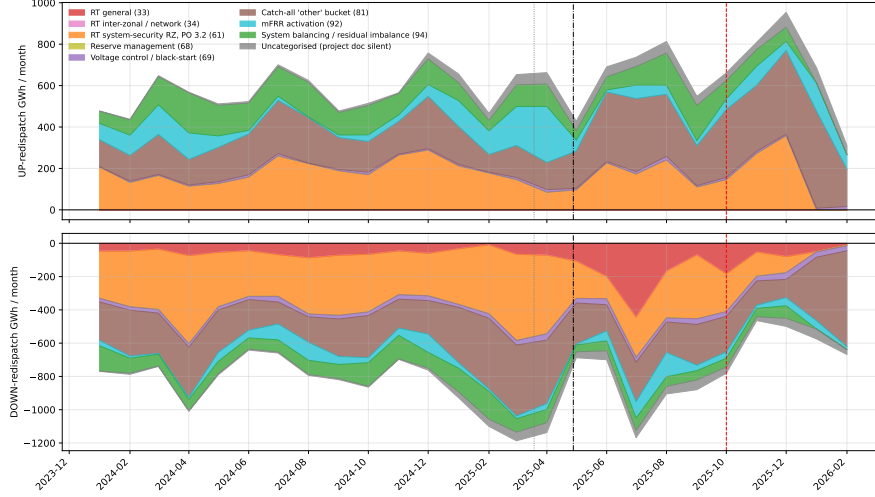


Figure 4: Full REE post-clearing intervention by *tipo redespacho* code, ESIOS archive 28 (totalrp48preccierre). Top: UP-redispach GWh / month; bottom: DOWN-redispach (negative axis). Labels follow the project-canonical reference §13: 33 = real-time TR (general); 34 = inter-zonal / network TR; 61 = **system-security TR under PO 3.2 (the post-IDA / real-time channel)**; 68 = reserve management; 69 = voltage control / black-start; 81 = catch-all “other”; 92 = mFRR activation; 94 = system balancing. Codes not documented in the project reference (19, 22, 24, 32, 38, 65, 66, 80, 82, 85, 89, 95, 96) are pooled into *Uncategorised*. Code 23 (~10.8 TWh up over 2024-2026, up-only) is excluded from this delivered-energy view because its profile is consistent with a capacity-reservation series (MW × hours, PO 7.2 banda-secundaria); stacking it on the same axis as activated-energy channels would be apples-to-oranges.

7.2 Per-firm × per-technology Path A reading

Programs: PHF – PDBF, per (firm, technology, month). Pre window 2024-01 → 2025-03; post window 2025-05 → 2026-01 (*operación reforzada*). Bilaterals cancel out (in both terms at the same MW); the residual is IDA + REE post-IDA RT.

Table 9: Post-DA intervention by firm and technology, GWh / month. Programs: PHF – PDBF, same construction as Table ?? but resolved per firm. The bilateral channel (PDBF – PDBC) is small for CCGT (Table ??); does *not* drive this gap.

Tech	Firm	Pre (GWh/mo)	Post (GWh/mo)	Δ (GWh/mo)	Ratio
CCGT	GE	213	413	+200	1.94
	GN	397	932	+534	2.34
	HC	42	94	+52	2.25
	IB	146	316	+169	2.16
Hydro	GE	-58	-253	-195	4.33
	GN	5	-33	-38	-6.96
	HC	-0	-11	-11	40.05
	IB	33	-84	-117	-2.55
Hydro_pump	GE	2	-15	-17	-7.14
	GN	4	3	-1	0.70
	IB	23	-1	-24	-0.03
Nuclear	GE	241	23	-218	0.10
	IB	1	-137	-138	–
Wind	GN	-9	-36	-27	4.02
	HC	-13	-50	-36	3.71
	IB	8	-81	-89	-10.63
Solar PV	GE	-0	-0	-0	9.13
	GN	-3	-11	-8	3.29
	HC	0	1	+1	6.26
	IB	3	-42	-45	-12.10

Reading. Every CCGT firm doubles its post-DA gap (GN 534 GWh/mo, the largest); Wind / Solar / Hydro / Nuclear mirror-decrease (REE substitutes CCGT for clean tech). Among Figure ??’s codes, only code 61 (PO 3.2 system-security TR) is *both* pay-as-bid *and* local-grid binding — the combination that yields zonal market power. Critical hours carry 3–5 $\times$ the per-day gap of flat or midday.

7.3 Per-firm CCGT post-clearing revenue (proxy)

Each (firm, tech, month) intervention multiplied by the regime-mean Fase I-up pay-as-bid price (ESIOS id 705) gives a post-clearing redispatch revenue proxy (aggregates Fase I+II + TR up).

Table 10: Per-firm CCGT up-redispatch revenue (proxy), EUR million per regime window. Big-4 sum in DA15/ID15 ($\sim 1,390\text{M}$) is close to the REE-published Fase I-up cost (1,200M in Table ??); the $\sim 15\%$ overshoot reflects TR-up and Fase II-up inclusion.

Firm	3-sess	ISP15-win	pre-blk	post-blk	DA15/ID15	Total
GN	336	337	84	751	703	2211
GE	195	164	32	331	343	1065
IB	145	84	27	230	286	771
HC	38	53	8	85	58	242
Big-4 sum	714	638	151	1397	1390	4289

Reading. **GN captures $\sim 50\%$ of Big-4 CCGT post-clearing revenue in every regime.** GN CCGT: €336M (3-sess) $\rightarrow$ €337M (ISP15-win) $\rightarrow$ €751M (post-blk) $\rightarrow$ €703M

(DA15/ID15) — $\sim 2\text{M/day}$ pre-reforzada, $\sim 5\text{--}8\text{M/day}$ post — concentrated in GN’s southern-corridor zones (Sur/Levante/Cataluña; Table ??).

7.4 CCGT marginal cost and the Fase I scarcity rent

We compute a bottom-up CCGT marginal cost as $MC = P_{\text{gas}}/\eta + P_{\text{CO}_2} \cdot EF/\eta + \text{VOM}$ with $\eta = 0.55$ LHV, $EF = 0.20$ tCO₂/MWh-thermal, $\text{VOM} = 4$ EUR/MWh-elec; gas from ESIOS id 1940, CO₂ from id 1391. Start-up/no-load adjustments shave ~ 10 EUR/MWh off the premium below.

Table 11: Approximate CCGT marginal cost vs market and Fase I prices, per regime (EUR/MWh-elec). Premium = Fase I up – MC_{central} before the ~ 10 EUR/MWh start-up adjustment.

Regime	DA spot	Fase I up (id 705)	MC _{central}	Premium
3-sess (Jun–Nov 24)	77	159	$\sim 100\text{--}115$	44 to 59
ISP15-win (Dec24–Mar25)	118	187	$\sim 125\text{--}150$	37 to 62
DA60/ID15 pre-blk	54	145	~ 130	15
DA60/ID15 post-blk	63	165	$\sim 100\text{--}110$	55 to 65
DA15/ID15	76	168	$\sim 100\text{--}110$	58 to 68

Reading. Fase I-up exceeds bottom-up CCGT MC by 40–70 EUR/MWh in every regime: a **30–60% mark-up over short-run MC** (after start-up adjustment) captured pay-as-bid by zonally-dominant CCGTs. DA spot in 2024–26 sits at or below CCGT MC much of the time. Scope: the units that clear Fase I, disproportionately the zonally-constrained corridor CCGTs ($\sim 20\text{--}30\%$ of the fleet, Table ??).

8 Geographic CCGT concentration and zonal market power

Why this matters. PO 3.2 redispatches relieve transmission-grid constraints *locally*, and the resulting redispatch energy is paid pay-as-bid (§??). The zonal-dominant CCGT operator therefore holds local market power on the Fase I / TR bids in its zone, independent of the EUPHEMIA day-ahead clearing. The fewer the eligible CCGTs in a constrained zone, the higher the resulting Fase I price.

Variable. CCGT installed capacity (MW) per (zone, firm), with HHI computed on capacity shares within each zone ($\times 10,000$). **Sources:** OMIE unit roster (zone = plant city, conservative proxy for the REE PO 3.2 zonal book) joined with ESIOS archive 110 (master data on registered installations).

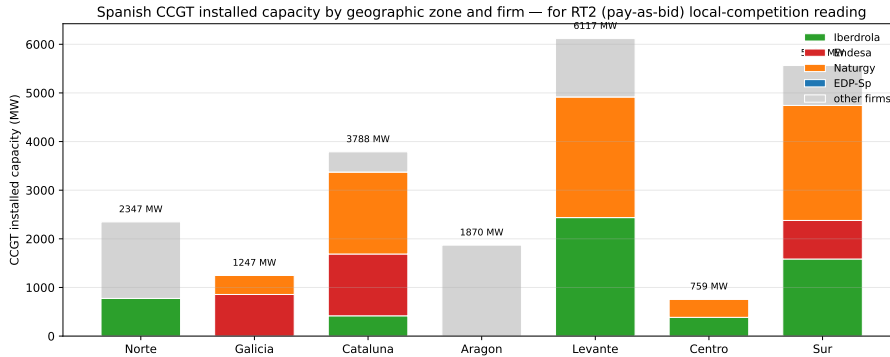


Figure 5: Spanish CCGT installed capacity (MW) by zone and firm. Bars stack each firm’s MW; zone labels (Sur, Levante, etc.) are grouped from plant-city assignments. Galicia and Centro are duopolies; Sur / Levante / Cataluña have wider firm diversity but GN holds the largest share in each.

Table 12: Per-zone CCGT capacity concentration. HHI on within-zone capacity shares ($\times 10,000$).

Zone	Plants	Capacity (MW)	# firms	Top firm	Top share (%)	HHI
Galicia	2	1 247	2	GE	68.6	5 695
Norte	10	2 347	2*	non-Big-4*	67.0	5 577
Aragón	3	1 870	2*	non-Big-4*	57.7	5 119
Centro	2	759	2	IB	50.9	5 001
Levante	12	6 117	3	GN	40.6	3 616
Cataluña	8	3 788	4	GN	44.5	3 348
Sur	13	5 563	4	GN	42.5	3 040

* Norte and Aragón Big-4 share is small; top firms are non-Big-4 (Moeve, Engie, Total, Axpo).

Reading. Galicia (HHI 5,695, 2 plants, GE 69%) and Centro (HHI 5,001, 2 plants, IB 51%/GN 49%) are CCGT duopolies. The three southern-corridor zones — Cataluña, Levante, Sur (HHI 3,000–3,600) — have GN at 40–45% and bear the heaviest reforzada post-blackout. Norte and Aragón are non-Big-4 dominated (Moeve, Engie, Total, Axpo).

8.1 Bid-side heterogeneity by zonal dominance

We split each CCGT unit into “dominant in its zone” (largest within-zone capacity share) vs “non-dominant” and compare the median DA sell-bid price by firm.

Table 13: Median DA sell-bid price by firm and zonal dominance, EUR/MWh. Per (unit-day) median; Pre = 2024 same-calendar, Post = 2025-05+ reforzada. Unit-counts (dominant / non-dominant): GN 15/2, GE 1/4, IB 1/9, HC 0/2.

Firm	Pre (2024 same-cal)				Post (2025-05+)			
	Dom	Non-dom	Diff	ratio	Dom	Non-dom	Diff	ratio
GN	844	866	−22	0.97	840	852	−12	0.99
GE	2,680	2,381	299	1.13	1,789	445	1,344	4.02
IB	156	170	−14	0.92	147	196	−49	0.75
HC	—	156	—	—	—	163	—	—

Reading. **GN**: cap-padded ~ 840 /MWh ladder applied *uniformly* across zones and regimes — fleet-wide Cournot-on-quantity policy. **GE**: zone-specific extensive-margin withholding — PGR5 in Galicia is fully parked, BES3/BES5/COL4 collapsed from parked to competitive post-reform. **IB** and **HC**: bid €150–220/MWh near short-run marginal cost (§??); IB shows no dominant-vs-non-dominant differential. The CNMC’s 2019 Naturgy sanction (§??) covered exactly the GN southern-corridor plants whose flat €840 ladder appears above.

8.2 Spatial test: did MTU15 swings predict the post-blackout restriction burden?

CNMC’s blackout report flags MTU15 as causing “transitions between negative and positive prices [...] leading to strong changes in production volumes in certain zones”. We test bigger MTU15 within-hour swings $\rightarrow$ bigger post-blackout restrictions two ways. (a) **Per-zone BSTS for CCGT** on daily CCGT PHF – PDBF (cutover 2025-04-28, Spec A wind+solar+gas co-variates + weekly cycle internally absorb seasonality). (b) **Per-unit cross-section** on all dispatchables (CCGT/Hydro/Pump), regressing the post-minus-pre PHF – PDBF jump on the unit’s within-hour SD with tech FE; *seasonality is differenced out by same-calendar matching*

(pre = May–Dec 2024, post = May–Dec 2025, same 8 months one year apart). Volatility score in both: per-unit within-hour SD of MW in the 40-day MTU15-IDA pre-blackout window (PHF).

Table 14: Spatial test results. (a) Per-zone BSTS for CCGT: AbsEffect on daily PHF – PDBF (GWh/day), pre 2024-06-14 to 2025-04-27, post 2025-04-28 to 2026-01-20. (b) Per-unit Spearman correlation of (post-pre) PHF – PDBF jump with the unit’s MTU15-IDA pre-blk within-hour MW SD, within tech.

(a) Per-zone CCGT BSTS				(b) Per-unit, within tech		
Zone	n	vol (MW)	BSTS effect	Tech	n	Spearman $_{\rho}$
Sur	9	4.09	15.29	CCGT	33	−0.17
Levante	9	4.09	5.47	Hydro	51	−0.57
Cataluña	5	1.70	4.97	Hydro_pump	8	−0.86
Norte	9	4.52	3.93	Pooled (tech FE): $\beta = -0.046$ ($p < 0.001$)		
Aragón	2	18.75	3.37			
Galicia	2	1.51	1.35			
Centro	2	3.65	0.75			
Cross-zone correlation (a): Pearson −0.08, Spearman 0.23. Bold $\equiv$ 95% CI excludes zero / $p < 0.01$.						

Reading. (i) The restriction jump is strongly heterogeneous across zones (Sur 15.3 vs Centro 0.75 GWh/day) — the zonal dimension of reforzada is real — and tracks the zonal-dominance map (§??): the four highest-effect zones sit in the southern-Mediterranean voltage-fragile corridor. (ii) But *volatility does not predict restriction burden in any tech* (after same-calendar seasonality control): the cross-zone CCGT correlation is weakly positive (0.23); the within-tech per-unit correlations for hydro and pump are sharply *negative* (−0.57, −0.86); the pooled tech-FE slope is −0.046 ($p < 0.001$). The CNMC’s zone-specificity is empirically supported; their proposed channel (MTU15 swings drive restrictions) is falsified — if anything, REE restricts the units that were *steadier* under MTU15-IDA, suggesting the binding constraint is grid topology / voltage criticality, not within-hour swing magnitude. Reproduction in [scripts/analysis/regulatory/](#).

9 System cost cascade: the full ajuste view

Every MWh cleared in DA/IDA triggers downstream settlements in ≥ 6 ancillary processes; the cascade is the falsifier for whether MTU15 reduces total system cost. The two pay-as-bid legs (Fase I and TR) are where local market power can bite; the marginal-priced balancing services (aFRR PICASSO, mFRR MARI, RR TERRE) do not.

9.1 Per-regime cost composition (REE-published cost indicators)

Total cost (EUR million) per regime, summed across ISP cells. **Sources:** REE-published cost indicators 1373/1374 (Fase I up/dn), 1375/1376 (Fase II up/dn), 1723/1724 (TR up/dn), 712/2127 (aFRR reserve up/dn), 726/727 (imbalance excess/deficit).

Table 15: Per-regime ajuste-cascade cost composition. EUR *million* per regime, summed across ISP-15 cells. Negative TR (dn) values reflect REE paying providers to come down. Per-day evolution of the same components in Figure ??.

cat regime	Fase I (dn)	Fase I (up)	Fase II	Imbalance	TR (dn)	TR (up)	aFRR reserve	Total
Pre-IDA (Jan-Jun 24)	10	822	141	0	0	501	208	1682
3-sess (Jun-Nov 24)	70	843	313	0	-4	668	273	2163
ISP15-win (Dec24-Mar25)	33	573	239	0	-1	627	264	1735
DA60/ID15 pre-blk	1	305	30	0	-17	148	53	520
DA60/ID15 post-blk	110	1625	425	0	-66	452	195	2741
DA15/ID15 (Oct-Dec 25)	44	1200	431	0	-14	170	113	1944

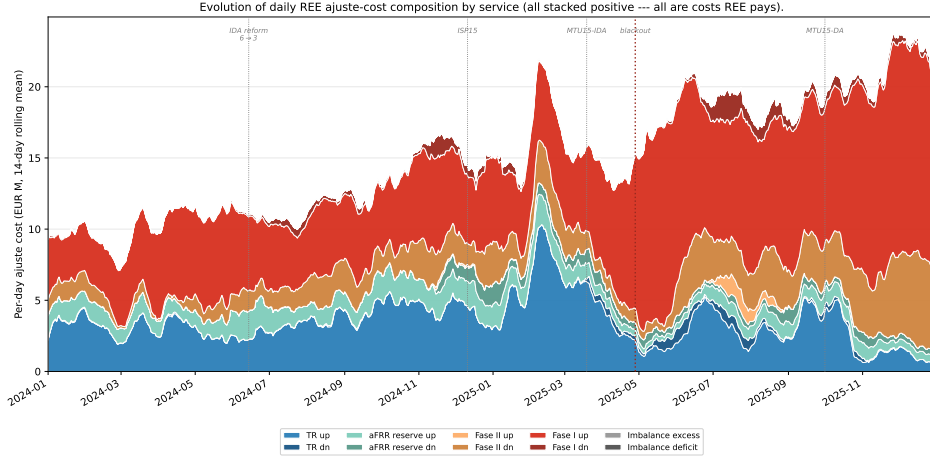


Figure 6: Daily REE ajuste-cost composition by service (14-day rolling mean), stacked positive. All components are gross costs REE pays providers; UP and DN flows are stacked together with no sign convention. Vertical dotted lines mark the five reform dates (IDA reform 2024-06-14, ISP15 2024-12-11, MTU15-IDA 2025-03-19, blackout 2025-04-28, MTU15-DA 2025-10-01). The TR (up) channel falls sharply after MTU15-IDA, the granularity efficiency channel; Fase I (up) balloons after the blackout, the reforzada-driven CCGT recall. Per-regime aggregates are in Table ??.

9.2 Per-day equivalents (the headline)

	3-sess	ISP15-win	DA60/ID15 pre-blk	DA60/ID15 post-blk	DA15/ID15
Length (days)	~170	~108	40	~155	92
Total cost (€M)	2 163	1 735	520	2 741	1 944
Per day (€M)	12.7	16.1	13.0	17.7	21.1
Fase I up / day	5.0	5.3	7.6	10.5	13.0
TR up / day	3.9	5.8	3.7	2.9	1.8

Raw per-day rates from Table ?. Regime windows differ in calendar mix, so the row-by-row deltas mix granularity, reforzada, and seasonality; for a per-day evolution that lets the eye separate the reform-driven inflection from the seasonal noise, see Figure ??.

Reading. Total ajuste cost €16.1M → €21.1M/day (31%) from ISP15-win to DA15/ID15: cost *redistributed* from TR (−4.0M/day; TR price halved €545 → €226/MWh as MTU15-IDA absorbed intra-hour scarcity) to Fase I (5.7M/day; price barely moved €187 → €168 but cost more than doubled). MTU15 is *not* purely cost-saving; the ~21M/day ajuste bill dwarfs any DA-auction margin. The TR price halving aligns with the IDA dual-direction surge documented in §??: finer-grained intraday bidding eroded the intra-hour scarcity that previously priced TR.

9.3 Robustness panels

Balancing stack. aFRR offer curve (ESIOS `curvas_ofertas_afrr`) and mFRR aggregated bids (ENTSO-E) show marginal-priced balancing is not concentration-distorted by the reforms (60–90 price steps per ISP, ~15% MW growth, aFRR-Down price-range expansion €115 → 384 tracking the Fase I-dn growth in Table ?; `scripts/analysis/balancing/balancing_stack_per_regime.py`).

Cross-border price gap. France is not a clean control for ID15 or DA15 (EU-wide SIDC/SDAC) but IS for Spain-specific ISP15 / reforzada. The Spain–France hourly DA-price gap widens against Spain by ~17 EUR/MWh across the 40-day MTU15-IDA pre-blackout window — ID15 is Spain-specific in magnitude despite the continental SIDC reform — and shifts 37 EUR/MWh at the reforzada onset (the Fase I rent at the price level). Reproduction: `scripts/analysis/regulatory/cross_b` quantitative detail in *preliminary results* §4.A.1 footnote.

10 Cournot reading and CNMC enforcement precedent

Table 16: CNMC sanction resolutions relevant to the practice. SNC-DE-175-17 covers Besós 4, PBCN 1/2, Sagunto 1/2/3, Málaga 1, SROQ 1 — the Cataluña / Levante / Sur zonal map (§??); CNMC benefit estimate €13.0 M over 4 months. LSE = Ley del Sector Eléctrico 24/2013.

Year	Firm / Plant	Expediente	LSE article	Sanction
2015	Iberdrola hydro	SNC-DE-046-14	64 muy grave (manipulation)	25.0 M EUR
2019	Naturgy CCGT	SNC-DE-175-17	65.34 grave (abnormal DA offers)	~7 M EUR
2019	Endesa Besós 3	SNC-DE-174-17	65.34 grave (abnormal DA offers)	6.6 M EUR
2023	Naturgy Sabón 3 (Galicia)	—	65.33 grave (RT abuse)	confidential
2023	Engie Castelnou	SNC-DE-152-22	grave (availability)	—
2023	Ignis Escatrón 2	SNC-DE-151-22	grave (availability)	—

GN’s near-constant $\sim 840/\text{MWh}$ DA ladder (Table ??) with quantity-side adjustment is the signature of Cournot-on-quantity — price-raising in DA dressed as “my offer price has not changed,” with rent captured in pay-as-bid Fase I (§??). The CCGT σ_p widening of *preliminary results* §A.4 sits on the in-band competing tier, not on this parked ladder. The 2019 Naturgy sanction (SNC-DE-175-17) is the precedent; 2024–26 is the same at fleet scale, with granularity as an extra tool.³

11 Do unit-level arbitrageurs exist? Buy and sell offers from production plants

The theoretical model in *model.tex* posits an arbitrageur agent that links the day-ahead (DA) and intraday auction (IDA) prices via a position s . Empirically, the natural signature of unit-level arbitrage is that a *production* unit (one that generates electricity) submits not only sell offers (the default for a producer) but also buy offers, i.e. repurchases part of an earlier sell-side commitment. We test this directly in OMIE offer-header data (`cab` for DA, `icab` for IDA, 2018-01-01 to 2026-01-31). For each unit-day (DA) or unit-session (IDA), we record whether at least one sell offer (`buy_sell = 'V'`) and at least one buy offer (`buy_sell = 'C'`) appear; we then aggregate by tech-group and report the share of seller unit-days/sessions that also carry a buy offer (Table ??).

Table 17: Production-unit buy-and-sell offers in OMIE auctions, by tech-group, 2018-01-01 to 2026-01-31. “% with both” is the share of unit-days (DA) or unit-sessions (IDA) on which the unit submitted at least one sell offer that *also* carried at least one buy offer. The DA market is pure sell-side for production: no unit ever submits a DA buy offer. IDA flips: every tech-group has non-trivial dual-direction bidding, with variable RES the most extreme.

Tech-group	n unit-days (DA)	% with both (DA)	n unit-sessions (IDA)	% with both (IDA)
Solar PV	944,918	0.00	2,214,353	30.39
Wind	537,764	0.00	2,389,721	17.62
Biomass	149,641	0.00	116,343	7.23
Hydro	119,357	0.00	347,350	6.37
Hydro_pump	21,533	0.00	54,949	5.09
Nuclear	42,769	0.00	13,799	4.44
Hydro.RES	311,347	0.00	446,228	4.38
Cogen	511,140	0.00	345,407	4.33
CCGT	137,481	0.00	271,139	4.25
Coal	5,320	0.00	7,119	0.81

³The ~ 56 April-2026 voltage-control expedientes (Art 65.8) are a separate regulatory channel.

Two facts stand out. First, the DA market is purely sell-side for production: across more than 2.7 million unit-days and ten tech-groups we observe *zero* DA buy offers from any production unit. This is consistent with DA being settled as a single hourly product per unit per day — there is no within-day price spread for a producer to exploit by buying back its own forward position, and the EUPHEMIA-style auction does not naturally accommodate units submitting both directions. Second, the IDA picture is the opposite: every tech-group has at least some unit-sessions with dual-direction bidding. Variable renewable sources (Solar PV at 30%, Wind at 18%) dominate — consistent with forecast-error rebalancing, where the operator buys back an earlier sell when the realised production forecast tightens. Among dispatchable techs, CCGT (4.3%), Hydro_pump (5.1%), Nuclear (4.4%) and Hydro (6.4%) all show non-zero shares; only Coal, with a near-zero share (0.81%), behaves as a pure seller.

The reading consistent with the model is the following. The DA’s auction structure (single price per unit per hour, EUPHEMIA-style) leaves no within-day spread for a producer to arbitrage in DA itself; arbitrage between DA and IDA is therefore observed not in the DA bid file but in the IDA bid file, where the producer rebalances. Variable-RES techs do most of this rebalancing because their forecast updates are largest; thermal and Hydro_pump do less but the share is not zero, which is direct evidence that the arbitrageur agent in the model is empirically active. Moreover, the share is bounded in the IDA — at most $\sim 30\%$ even for Solar PV — consistent with capacity-binding limited arbitrage (Result 3 of Ito and Reguant, 2016), not the unlimited-strategic regime in which every product would carry an arbitrage position.

11.1 Measure construction: what “arbitrage intensity” captures and what it doesn’t

The measure used in this section requires careful documentation because it is not a direct observation of arbitrage profit, capital, or net position. We construct it as follows:

Definition. For each calendar date d , IDA session s , and production unit u , we record whether the unit submitted at least one sell offer (`buy_sell = 'V'`) and at least one buy offer (`buy_sell = 'C'`) in the IDA offer-header file `icab.all`. A unit-session is a *dual-direction* bid if both directions appear. Our daily arbitrage-intensity outcome is the per-tech-group count of dual-direction unit-sessions on date d , summed across the three IDA sessions of the day.

What the measure captures. A unit-session with both buy and sell offers is consistent with three economically distinct behaviours, all of which contain an arbitrage component in the broad sense: (i) *position rebalancing*, where the unit’s DA-side commitment exceeds (or falls short of) its updated production forecast and the unit submits a buy (sell) to close the gap; (ii) *pure financial arbitrage*, where the unit buys at a low spot price expecting to sell at a higher spot price (or to net out against its DA position) without any underlying production change; and (iii) *strategic withholding then re-clearing*, where the unit deliberately under-offers at DA expecting to capture a higher IDA clearing price, then re-buys some of the previously-sold MW. All three are forms of unit-level arbitrage between the DA-cleared price and the IDA-cleared price.

What the measure does NOT capture. The measure is binary at the unit-session level: it does not encode the *volume* of dual-direction offers (a 1 MW buy and a 100 MW sell count the same as a 50/50 split), the *price-spread* between the buy and sell offers, or the *net* cleared MWh direction across the session. It also does not capture cross-session arbitrage (selling in IDA-1, buying in IDA-2), within-day arbitrage of the same unit’s IDA position, or arbitrage between IDA and the continuous-intraday market XBID. Within-session netting between sell and buy offers can occur in our measure even when total unit MWh stays unchanged, so high dual-direction counts can co-exist with low absolute net repositioning. Conversely, a unit shifting its entire 100 MW sell from session 1 to session 2 (no buys) is genuine arbitrage that our measure misses entirely.

Sources of bias for the model interpretation. The model’s arbitrageur agent is profit-motivated (carries position s to exploit the forward-spot wedge). Our measure conflates this with two innocent behaviours: forecast updates (especially for RES, which see large within-hour forecast revisions) and operational re-positioning around physical constraints. The Solar PV 30% share probably reflects forecast updates more than profit-driven arbitrage; the much lower CCGT/Nuclear shares ($\sim 4\%$) are closer to the model’s intended profit-arbitrage activity, since thermal units have small forecast errors and minimal physical re-positioning needs. The measure should therefore be read as a *lower* bound on the share of unit-sessions with any arbitrage component, not as a measure of pure profit-arbitrage intensity.

Implication for the model’s P7 prediction. The cross-tech heterogeneity discussed below should be read against this background: the measure is well-suited to detect changes in dual-direction prevalence, but the level interpretation requires care.

11.2 Did arbitrage intensity rise at MTU15-IDA? BSTS on daily dual-direction counts per tech

If the model’s arbitrageur agent is empirically active, MTU15-IDA (cutover 2025-03-19) should mechanically raise the per-tech daily dual-direction count: each clock-hour now hosts four matched 15-min IDA products against one DA product, so the per-clock-hour space for arbitrage is four times larger. We run a per-tech BSTS on the daily dual-direction count, with wind+solar+gas covariates (matching *preliminary results* Spec A), pre-window 2022-06-14 (post 6 $\rightarrow$ 3 IDA reform) to 2025-03-18, and the 40-day pre-blackout post-window 2025-03-19 to 2025-04-27. Clearly identifies the MTU15-IDA effect with *reforzada* held constant (*reforzada* onset is 2025-04-28).

Table 18: BSTS effect of MTU15-IDA on daily dual-direction unit-session count, by tech-group. Stars on p -values: $*p<0.10$, $**p<0.05$, $***p<0.01$. Pre mean and post mean are daily counts averaged over their respective windows; absolute effect is the BSTS counterfactual gap. The heterogeneity across techs is the headline read: forecast-driven techs (Solar PV, Hydro_RES, Biomass) jump sharply, while thermal and pumping techs do not.

Tech-group	Pre mean	Post mean	Abs. effect	Rel. effect	p
Solar PV	828.8	1100.2	+271.4***	+33.6%	<0.001
Hydro_RES	25.3	43.5	+18.1***	+76.3%	<0.001
Biomass	3.2	9.8	+6.6***	+521%	<0.001
Nuclear	0.4	2.0	+1.6***	+450%	<0.001
Wind	648.7	682.2	+33.5	+6.4%	0.32
Hydro	33.8	33.6	−0.1	+1.5%	0.48
Cogen	19.4	21.5	+2.1	+13.4%	0.22
CCGT	6.1	5.1	−1.0	−27.1%	0.44
Hydro_pump	5.6	4.4	−1.2*	−19.7%	0.05
Coal	0.5	0.1	−0.4***	−73.5%	0.001

The pattern is sharply stratified by tech. *Forecast-driven techs* (Solar PV, Hydro_RES, Biomass) show massive jumps in dual-direction count, with Solar PV alone adding ~ 270 dual-direction unit-sessions per day — consistent with the mechanism in §4.A.3 of the preliminary results: per-period IDA scheduling lets aggregators track their 15-min production forecast more closely, which mechanically requires more buy-then-sell rebalancing as the forecast updates within the hour. *Wind, Hydro and Cogen* are flat, plausibly because their dual-direction shares were already saturated before MTU15-IDA. *Thermal and storage* (CCGT, Coal, Hydro_pump) show no increase or a small decrease.

The CCGT and Hydro_pump results require an extra layer of caveat before any model interpretation. By the MTU15-IDA post-window (March–April 2025), Spanish solar was crowding the

merit order and CCGT day-ahead cleared volume had already collapsed to near zero (the q_1 drop in §??); a unit producing essentially nothing has very little to rebalance, so the CCGT BSTS null is largely a no-production result rather than a tested-and-rejected model prediction. The dual-direction count is a sensible *rebalancing* measure only for techs that are actually clearing volume in the post-window — which forecast-driven RES and biomass are (high cleared MW under MTU15-IDA), and which CCGT, Coal and Hydro_pump are not. Pure profit-arbitrage by thermal units may also operate through net-direction changes across sessions (a phenomenon the measure misses, per §??). The CCGT scale-up at DA15 documented in *preliminary results* §4.A.2 — which lands in a window where CCGT is producing again — is the more direct empirical signature of thermal arbitrage, through quantity migration into DA rather than within-session position rebalancing in IDA. *There is also a third arbitrage channel the within-OMIE dual-direction measure cannot see by construction:* during reforzada (continuous from 2025-04-28, overlapping all of the post-blackout window), the dominant CCGT arbitrage is between Fase I up-redispach (pay-as-bid, REE-paid) and the wholesale auction clear — the parked-tier / Fase I-recall pattern documented in §??. **Read together, the three findings characterise three distinct empirical arbitrage channels:** (a) *within-session forecast-driven rebalancing in IDA*, dominated by RES/biomass and amplified by MTU15-IDA; (b) *between-market quantity migration into DA*, dominated by CCGT and amplified by DA15; (c) *Fase I-vs-wholesale arbitrage under reforzada*, dominated by CCGT and driven by the post-blackout regulatory regime rather than by the granularity reforms.

11.3 What to elevate from this document into preliminary_results: selective glossary

The preliminary-results main memo carries the headline identification claims (Spec A BSTS, Spec B per-hour-class BSTS, Spec C per-curve DiD). The findings in this companion document deepen those headlines but most do not warrant elevation. Below is a deliberately short list of items that I believe should move into the preliminary memo — to its appendix unless flagged otherwise — because they discipline a model claim or close an identification loop the preliminary memo currently leaves open. Items not on this list stay here.

- **Production-unit arbitrageurs exist (§??)** → preliminary memo *appendix*. Directly validates the model’s arbitrageur agent (P7): production units submit zero DA buy offers but 4–30% dual-direction IDA unit-sessions across all tech groups. Worth a half-page appendix figure plus the per-tech table.
- **Dual-direction count jumps at MTU15-IDA for forecast-driven techs (§??)** → preliminary memo *appendix*. Direct identification of one arbitrage channel via BSTS. Pair with the §4.A.3 renewable-cleared-MW reallocation finding (which it complements, not duplicates).
- **Geographic CCGT concentration and zonal market power (§??)** → preliminary memo *main text* (one paragraph) or *appendix*. Provides the market-structure scaffolding for the §4.A.2 CCGT scale-up reading: the bidder is identifiably concentrated in specific zones, which makes the strategic-bidder identification of the model concrete in the data.
- **Cournot reading and CNMC enforcement precedent (§??)** → preliminary memo *main text* (one sentence) or footnote. The 2019 Naturgy sanction (SNC-DE-175-17) is the regulatory analogue of the model’s strategic-bidder withholding mechanism. Useful for the §4.A.2 “competitive clearing anticipates expected REE’s intervention” footnote.

The list is intentionally short: four items in total, two of which (the arbitrage findings) come from this session’s work and the other two of which (geographic concentration, CNMC precedent) discipline existing headline claims rather than introducing new ones. The bid-curve-shape

decompositions (§??), tech-mix by hour-class (§??), continuous-intraday repositioning (§??), q_1 CCGT drop (§??), REE post-clearing inclusion (§??), and full ajuste cascade (§??) all serve as supplements to existing preliminary-memo findings rather than as candidates for elevation in their own right.